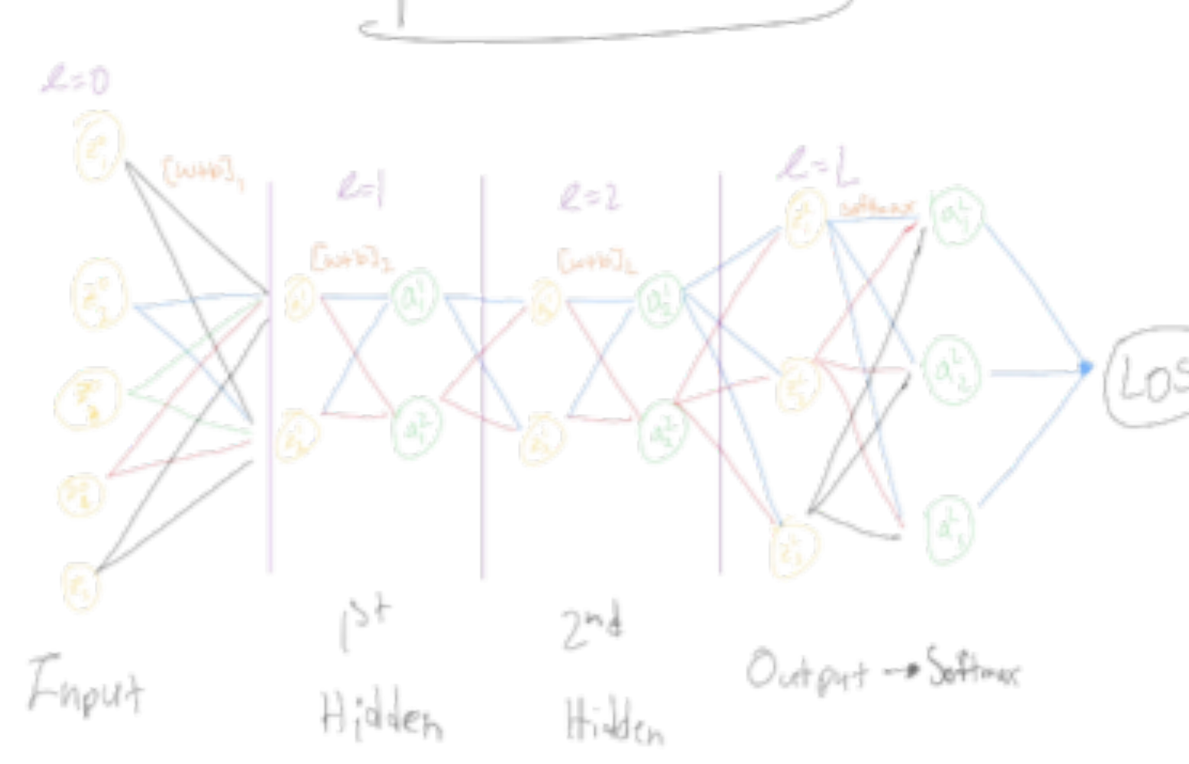
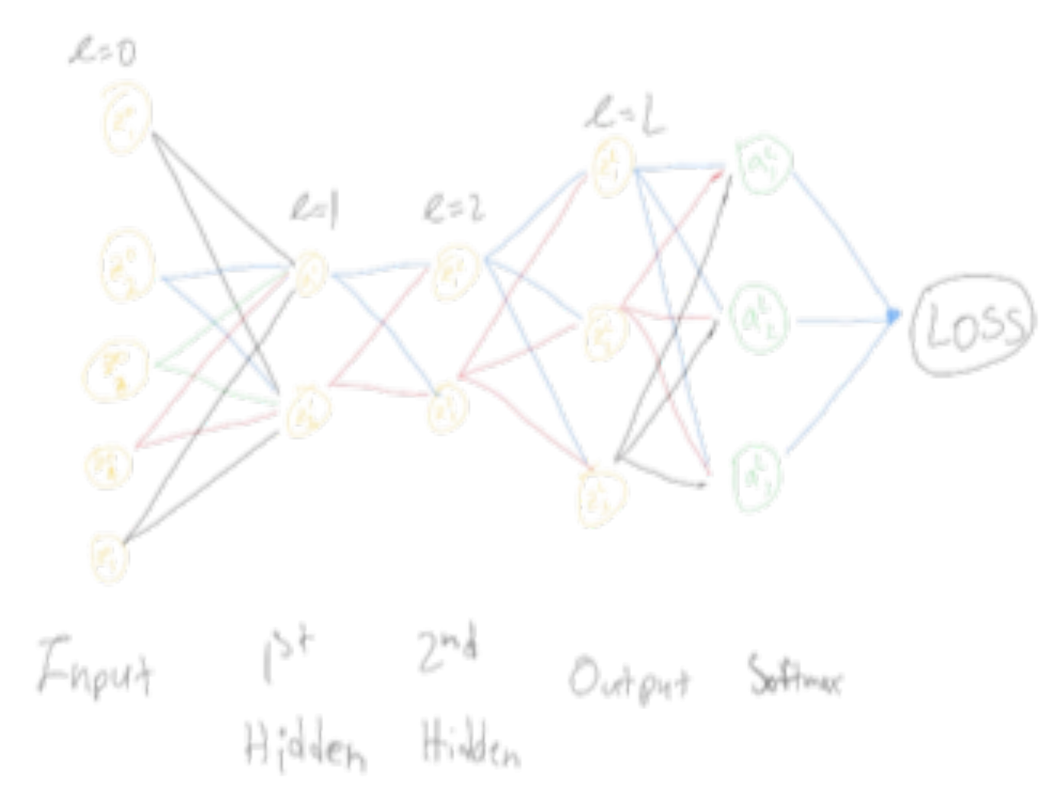
Back propagation

Example network



V1:



Notation & Set-Up

- ℓ: denotes specific layer #, with $\ell=1$ for 1st hidden layer & L = output layer
- W^{ℓ}_{jk}: Weight connecting neuron k in layer $\ell-1$ to neuron j in layer ℓ
- b^{ℓ}_{j}: Bias for neuron j in layer ℓ
- z^{ℓ}_{j}: Net weighted input for neuron j in layer ℓ
- a^{ℓ}_{j}: Activation for neuron j in layer ℓ (where $a^{\ell}_j = \sigma(z^{\ell}_j)$). For input layer, $a^0 = x$
- δ^{ℓ}_{j}: Error gradient for neuron j in layer ℓ, defined as $\frac{\partial L}{\partial z^{\ell}_j}$
- N^{ℓ}: # of neurons in layer ℓ

Error Gradient for Output Layer

Multiclass - Classification Loss Function:

1.)

a.) $\text{Loss }(L) = -\sum_{c=1}^{C} y_c \cdot \ln(a^L_c)$

where y_c = one-hot encoded ground truth at class c

and a^L_c = Softmax activation of neuron/class c in layer L

b.) $a^L_i = \frac{e^{(z^L_i)}}{\sum_{m=1}^{C} e^{(z^L_m)}}$ => (So~~f~~tmax)

2.) $\delta^L_j = \frac{\partial L}{\partial z^L_j} = \underline{\quad}$ (completely foiled out)

But, more succintly:

$$\delta^L_j = \sum_{c=1}^{C} \frac{\partial L}{\partial a^L_c} \cdot \frac{\partial a^L_c}{\partial z^L_j},$$

where there are 2 cases for when $j=c$ & $j \neq c$. This makes sense because the softmax activation a^L_c is influenced by its respective weighted neuron z^L_c more than a different neuron $z^L_{j\neq c}$ because the numerator of a^L_c contains a function of z^L_c, but not of $z^L_{j\neq c}$.

$$\delta^L_j = \left[\frac{\partial L}{\partial a^L_j} \cdot \frac{\partial a^L_j}{\partial z^L_j}\right] + \left[\sum_{c\neq j}^{C} \frac{\partial L}{\partial a^L_c} \cdot \frac{\partial a^L_c}{\partial z^L_j}\right]$$

$$= \left[\frac{-y_j}{a^L_j} \cdot \left(a^L_j \cdot (1-a^L_j)\right)\right] + \left[\sum_{c\neq j}^{C} \frac{-y_c}{a^L_c} \cdot (-a^L_c \cdot a^L_j)\right]$$

$$= \left[-y_j \cdot (1-a^L_j)\right] + \left[\sum_{c\neq j}^{C} y_c \cdot a^L_j\right]$$

$$= -y_j + \underbrace{y_j a^L_j + \sum_{c\neq j}^{C} y_c \cdot a^L_j}_{=\text{ sum of all } c \text{ from } 1 \text{ to } C} \quad = -y_j + \sum_{c=1}^{C} y_c a^L_j$$

$$= -y_j + a^L_j \cdot \underbrace{\left(\sum_{c=1}^{C} y_c\right)}_{=1 \text{ (sum of one-hot encoded vector } = 1)} = \boxed{a_j - y_j}$$

For every neuron j in the output layer, $\delta^L_j = a_j - y_j$. Thus, the error gradient vector:

$$\delta^L = \hat{a}_j - \hat{y}_j$$

Foiled Out

$$\text{loss} = -\sum_{i=1}^{C} y_i \cdot \ln(a^L_i)$$

$$= -\left[(y_1 \ln(a^L_1)) + (y_2 \ln(a^L_2)) + (y_3 \ln(a^L_3))\right]$$

$$\frac{\partial L}{\partial z^L_1} = \left(\frac{\partial L}{\partial a^L_1} \cdot \frac{\partial a^L_1}{\partial z^L_1}\right) + \left(\frac{\partial L}{\partial a^L_2} \cdot \frac{\partial a^L_2}{\partial z^L_1}\right) + \left(\frac{\partial L}{\partial a^L_3} \cdot \frac{\partial a^L_3}{\partial z^L_1}\right)$$

$$= \left[-\frac{y_1}{a^L_1} \cdot \overbrace{\left(\frac{\partial}{\partial z^L_1}\sigma(a^L_1)\right)}^{a}\right] + \left[-\frac{y_2}{a^L_2} \cdot \overbrace{\left(\frac{\partial}{\partial z^L_1}\sigma(a^L_2)\right)}^{b}\right] + \left[-\frac{y_3}{a^L_3} \cdot \overbrace{\left(\frac{\partial}{\partial z^L_1}\sigma(a^L_3)\right)}^{c}\right]$$

a.) $\frac{\partial}{\partial z^L_1}\sigma(a^L_1) = \frac{\left(e^{(z^L_1)} \cdot \sum_{m=1}^{C} e^{(z^L_m)}\right) - \left(e^{(z^L_1)} \cdot e^{(z^L_1)}\right)}{\left(\sum_{m=1}^{C} e^{(z^L_m)}\right)^2} = \left(\frac{e^{(z^L_1)}}{\sum_{m=1}^{C} e^{(z^L_m)}}\right) - \left(\frac{e^{(z^L_1)}}{\sum_{m=1}^{C} e^{(z^L_m)}}\right)^2$

$= \frac{u'v - uv'}{v^2}$

$f = e^{(z^L_1)}$

$g = \sum_{m=1}^{C} e^{(z^L_m)}$

$\frac{\partial}{\partial z^L_1} f = e^{(z^L_1)}$

$\frac{\partial}{\partial z^L_1} g = e^{(z^L_1)}$

$= a^L_1 - (a^L_1)^2$

$= \boxed{a^L_1(1-a^L_1)}$

b.) $\frac{\partial}{\partial z^L_1}\sigma(a^L_2) = \frac{\cancel{0} - \left(e^{(z^L_2)} \cdot e^{(z^L_1)}\right)}{\left(\sum_{m=1}^{C} e^{(z^L_m)}\right)^2} = -\left[\frac{e^{(z^L_2)}}{\sum_{m=1}^{C} e^{(z^L_m)}} \cdot \frac{e^{(z^L_1)}}{\sum_{m=1}^{C} e^{(z^L_m)}}\right]$

Identical for arbitrary $z_{i\neq c}$

$f = e^{(\overset{a_2}{z^L_2})}$

$g = \sum_{m=1}^{C} e^{(z^L_m)}$

$\frac{\partial}{\partial z^L_1} f = 0$

$\frac{\partial}{\partial z^L_1} g = e^{(z^L_1)}$

$= \boxed{-(a^L_2 \cdot a^L_1)}$

c.) $\frac{\partial}{\partial z^L_1}\sigma(a^L_3) = \frac{\cancel{0} - \left(e^{(z^L_3)} \cdot e^{(z^L_1)}\right)}{\left(\sum_{m=1}^{C} e^{(z^L_m)}\right)^2} = -\left[\frac{e^{(z^L_3)}}{\sum_{m=1}^{C} e^{(z^L_m)}} \cdot \frac{e^{(z^L_1)}}{\sum_{m=1}^{C} e^{(z^L_m)}}\right]$

$f = e^{(\overset{a_3}{z^L_3})}$

$g = \sum_{m=1}^{C} e^{(z^L_m)}$

$\frac{\partial}{\partial z^L_1} f = 0$

$\frac{\partial}{\partial z^L_1} g = e^{(z^L_1)}$

$= \boxed{-(a^L_3 \cdot a^L_1)}$

$$\frac{\partial L}{\partial z^2_1} = \left(\frac{\partial L}{\partial a^L_1} \cdot \left(-a^L_1(1-a^L_1)\right)\right) + \left(\frac{\partial L}{\partial a^L_2} \cdot \left(-(a^L_2 \cdot a^L_1)\right)\right) + \left(\frac{\partial L}{\partial a^L_3} \cdot \left(-(a^L_3 \cdot a^L_1)\right)\right)$$

$$= \left(\frac{-y_1}{\cancel{a^L_1}} \cdot \left(\cancel{a^L_1}(1-a^L_1)\right)\right) + \left(\frac{-y_2}{\cancel{a^L_2}} \cdot \left(-(\cancel{a^L_2} \cdot a^L_1)\right)\right) + \left(\frac{-y_3}{\cancel{a^L_3}} \cdot \left(-(\cancel{a^L_3} \cdot a^L_1)\right)\right)$$

$$= -y_1(1-a^L_1) + y_2 a^L_1 + y_3 a^L_1 = -y_1 + y_1 a^L_1 + y_2 a^L_1 + y_3 a^L_1$$

$$= -y_1 + a^L_1 \cdot \underbrace{(y_1 + y_2 + y_3)}_{=1} = -y_1 + a^L_1 = \boxed{a^L_1 - y_1}$$

Partial Derivative of Loss w.r.t. every Neuron

Formula: $z^{\ell}_j = \left[\sum_{k=1}^{N^{\ell-1}} (w^{\ell}_{jk} \cdot a^{\ell-1}_k)\right] + b^{\ell}_j$, **Find:** $\frac{\partial L}{\partial z^{\ell}_j}$

Start w/ case $\ell = L-1$:

☆ Find $\frac{\partial L}{\partial z^{L-1}_j} = \delta^{L-1}_j$

$$\delta^{L-1}_j = \frac{\partial L}{\partial z^{L-1}_j} = \sum_{k=1}^{N^L} \overbrace{\frac{\partial L}{\partial z^L_k}}^{=\delta^L_k} \cdot \frac{\partial z^L_k}{\partial z^{L-1}_j}$$

$z^L_k = \left[\sum_{j=1}^{N^{L-1}} (w^{L-1}_{kj} \cdot a^{L-1}_j)\right] + b^L_k$

$a^{L-1}_j = \sigma(z^{L-1}_j) \rightarrow z^L_k = \left[\sum_{j=1}^{N^{L-1}} (w^{L-1}_{kj} \cdot \sigma(z^{L-1}_j))\right] + b^L_k$

$\frac{\partial z^L_k}{\partial z^{L-1}_j} = (w^{L-1}_{kj} \cdot \sigma'(z^{L-1}_j))$

$$\delta^{L-1}_j = \sum_{k=1}^{N^L} \delta^L_k \cdot (w^{L-1}_{kj} \cdot \sigma'(z^{L-1}_j))$$

$$= \sum_{k=1}^{N^L} \delta^L_k \cdot w^{L-1}_{kj} \cdot \sigma'(z^{L-1}_j) = (\sigma'(z^{L-1}_j)) \cdot \left(\sum_{k=1}^{N^L} \delta^L_k \cdot w^{L-1}_{kj}\right)$$

$\delta^{L-1} = [\delta^{L-1}_{j=1}, \delta^{L-1}_{j=2}, \delta^{L-1}_{j=3}, \ldots \delta^{L-1}_{j=N^{L-1}}]$ => vector of each δ^{L-1}_j

$\left(\sum_{k=1}^{N^L} \delta^L_k \cdot w^{L-1}_{kj}\right)$:

1.) Shape (Post-Sum) = [1×1] (because it's a Σ, & also it = $\sigma'(z^{L-1}_j)$ shape)

2.) ☆ Assume W is of shape [output × input] = [k × j]

=> [1×k] · [k×1] = [1×1]

=> Either: $(\delta^L)(W^L_j)$ OR $(W^L_j)^T(\delta^L)$ ✓

=> When iterating over all values for j, we can simplify to:

$$\delta^{L-1} = \left((W^L)^T \delta^L\right) \odot \sigma'(\vec{z}^{L-1})$$

, where ⊙ is the element-wise (Hadamard) product (builds the vector δ^{L-1})

Generalize for all layers:

$$\delta^{\ell-1} = \left((W^{\ell})^T \delta^{\ell}\right) \odot \sigma'(\vec{z}^{\ell-1})$$

Derivative of Loss w.r.t. weights & biases

$$z^{\ell}_j = \left[\sum_{k=1}^{N^{\ell}} (w^{\ell}_{jk} \cdot a^{\ell-1}_k)\right] + b^{\ell}_j$$

$$\frac{\partial L}{\partial w^{\ell}_{jk}} = \frac{\partial L}{\partial z^{\ell}_j} \cdot \frac{\partial z^{\ell}_j}{\partial w^{\ell}_{jk}} = \delta^{\ell}_j \cdot a^{\ell-1}_k$$

$$\frac{\partial L}{\partial b^{\ell}_j} = \frac{\partial L}{\partial z^{\ell}_j} \cdot \frac{\partial z^{\ell}_j}{\partial b^{\ell}_j} = \delta^{\ell}_j$$

Updating weights & biases

α = Learning Rate

$$w^{\ell}_{jk} = w^{\ell}_{jk} - \left[\alpha \cdot (\delta^{\ell}_j \cdot a^{\ell-1}_k)\right]$$

$$b^{\ell}_j = b^{\ell}_j - \left[\alpha \cdot (\delta^{\ell}_j)\right]$$

↓ Matrix Multiplication

$$W^{\ell} = W^{\ell} - \left[\alpha \cdot (\hat{\delta}^{\ell} \cdot \hat{a}^{\ell-1})\right]$$

$$b^{\ell} = b^{\ell} - \left[\alpha \cdot \delta^{\ell}\right]$$